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Inventor(s)	Maruthamuthu; Saravana

Integrated circuit (IC) package employing on-package tunable inductor formed in redistribution layer (RDL) for impedance tuner circuit, and related methods

Abstract

Integrated circuit (IC) package employing on-package tunable inductor formed in redistribution layer (RDL) for impedance tuner circuit, and related methods. The IC package includes an impedance tuner circuit that includes a tunable inductor that can be tuned to change the frequency response of the impedance tuner circuit. To reduce the circuit area, the tunable inductor is formed in a RDL of a package substrate of the IC package. The IC package also includes a semiconductor die ("die") that includes other components of the impedance tuner circuit that are coupled to the tunable inductor by the die being coupled to the package substrate. In this manner, by the tunable inductor being formed in a RDL in the package substrate, the signal path lengths between the tunable inductor and other components of the tunable impedance circuit are reduced, thereby reducing inductance path resistance and improving quality (Q) factor of the tunable inductor.

Inventors:	**Maruthamuthu; Saravana (Bangalore, IN)**
Applicant:	**QUALCOMM Incorporated** (San Diego, CA)
Family ID:	**1000008779407**
Assignee:	**QUALCOMM INCORPORATED (San Diego, CA)**
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USPC: 257/531

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10490621	12/2018	Thadesar	N/A	H01L 23/49822
11451251	12/2021	Horita	N/A	H04B 1/1018
2016/0233192	12/2015	Dosluoglu	N/A	N/A
2017/0207022	12/2016	Song	N/A	H01L 23/49822
2017/0207293	12/2016	Song	N/A	H10D 1/20
2021/0160999	12/2020	Dogiamis et al.	N/A	N/A
2021/0313284	12/2020	Noori	N/A	H01L 25/072
2021/0313285	12/2020	Noori	N/A	H01L 23/49861
2021/0313935	12/2020	Noori	N/A	H01L 23/481

OTHER PUBLICATIONS

International Search Report and Written Opinion for International Patent Application No. PCT/US2023/021664, mailed Sep. 1, 2023, 18 pages. cited by applicant

Primary Examiner: Anya; Igwe U

Attorney, Agent or Firm: Withrow & Terranova, PLLC

Background/Summary

BACKGROUND

I. Field of the Disclosure

(1) The field of the disclosure relates to impedance tuners that can be employed in a transceiver circuit to filter radio-frequency (RF) signals, wherein the impedance tuners employ a tunable

resistor, capacitor, and inductor network to filter particular frequency bands.

II. Background

(2) A radio frequency (RF) transceiver includes a RF transmit circuit chain that conventionally includes an RF power amplifier (PA) to amplify a RF signal for transmission as an output RF signal within a given frequency band. The RF transmit circuit chain may also include an output impedance tuner circuit to filter the output RF signal in a desired frequency range or bandwidth. The RF transceiver also includes a RF receive circuit chain that conventionally includes a low-noise amplifier (LNA) coupled to an antenna that receives an input RF signal. The RF receiver circuit chain may also include an input impedance tuner circuit to filter the input RF signal to the desired frequency range of bandwidth. The impedance tuner circuits conventionally include a resistor, capacitor, and inductor network to filter the RF signals. The impedance tuner circuits may also include a switched variable capacitor and series of switches that can be programmed to selectively couple inductors into the RF signal path to control the filter performance.

(3) It may be desired to integrate a RF transceiver in an integrated circuit (IC) package that can be coupled to a printed circuit board (PCB) to conserve area. This is opposed to providing and coupling the discrete components of the RF transceiver on the PCB. However, discrete inductors for the impedance tuner are mounted on the PCB outside the IC package due to their size. These discrete inductors are coupled through programmable switches to the impedance tuner in the IC package. External pins of the IC package are coupled to electrical traces in the PCB coupled to the discrete inductors to the impedance tuner circuit in the IC package. However, the discrete inductors being located outside the IC package increases the length of the conductive paths coupling the discrete inductors to the impedance tuner circuit, thereby increasing resistance in the inductive paths. Increasing resistance in the inductive paths reduces the quality (Q) factor of the discrete inductors, thereby reducing the filter performance of the impedance tuner circuit.

SUMMARY OF THE DISCLOSURE

(4) Aspects disclosed herein include an integrated circuit (IC) package employing an on-package tunable inductor formed in a redistribution layer (RDL) for an impedance tuner circuit. Related methods are also disclosed. The IC package includes an impedance tuner circuit that includes a tunable inductor that can be tuned to change the frequency response of the impedance tuner circuit. To reduce circuit area needed to provide the tunable inductor, the tunable inductor is integrated in an IC package. In exemplary aspects, the tunable inductor is formed in a RDL of a package substrate of the IC package. The IC package also includes a semiconductor die (“die”) that includes other components of the impedance tuner circuit coupled to the tunable inductor by the die being coupled to the package substrate. In this manner, by the tunable inductor being formed in a RDL in the package substrate, the signal path lengths between the tunable inductor and other components of the tunable impedance circuit are reduced, thereby reducing the inductance path resistance and improving the quality (Q) factor of the tunable inductor. Also, by forming the tunable inductor in a RDL of the IC package, the tunable inductor can be formed of the desired geometry to provide the desired inductance frequency response.

(5) In exemplary aspects, the tunable inductor can be formed in the RDL of a package substrate of an IC package as a metal line having a first terminal and second terminal that can be coupled to a circuit to couple the inductance of the tunable inductor to the circuit. The tunable inductor is also formed in the RDL to have one or more shunt taps. A shunt tap is two metal lines formed in the tunable inductor that have ends adjacent to each other, but forming a short circuit. One or more switches in an active semiconductor layer in a front-end-of-line (FEOL) of the die as part of the impedance tuner circuit are coupled to respective shunt taps in the tunable inductor through signal path routing in the package substrate between the die and the tunable inductor in the RDL. The switches in the die can be programmed/controlled to either shunt the shunt tap or leave their respective, coupled shunt tap as an open circuit, to change the inductance of the inductor and shift the frequency response of the impedance tuner circuit. In this manner, the switches allow the

inductance of the inductor to be tuned through the controlling of the shunting or not shunting of the one or more shunt taps in the inductor. The tunable inductor being disposed in the RDL in the package substrate of the IC package locates the shunt taps closer to respective switches in the die to reduce inductance path length, thereby reducing resistance and improving inductance Q factor. Also, because the inductor is formed as a metal line in the RDL, the inductor and its shunt taps can be formed in the RDL of the desired geometry and connectivity arrangement as part of fabricating the package substrate, to provide the desired inductance tunability of the inductor.
(6) In other exemplary aspects, the tunable inductor is formed as a two-dimensional (2D) inductor such that the tunable inductor is formed in a single RDL in the package substrate of the IC package. In another exemplary aspect, the tunable inductor is formed as three-dimensional (3D) inductor such that the tunable inductor is formed in multiple RDLs of the package substrate of the IC package.
(7) In this regard, in one exemplary aspect, an IC package is provided. The IC package comprises an impedance tuner circuit comprising a first switch, and an inductor comprising a first terminal and a second terminal. The IC package also comprises a package substrate comprising a first RDL, the first RDL comprising the inductor. The IC package also comprises a die coupled to the package substrate, the die comprising the first switch. The inductor comprises a first metal line in the first RDL, the first metal line comprising the first terminal and the second terminal, and a first shunt tap in the first RDL. The first shunt tap is coupled to the first terminal and the second terminal. The first switch is coupled to the first shunt tap.
(8) In another exemplary aspect, a method of fabricating an IC package is provided. The method comprises forming a first switch in a die. The method also comprises forming an inductor in a package substrate, comprising forming a first metal line of the inductor in the first redistribution layer (RDL) in the package substrate, the first metal line comprising a first terminal and a second terminal, and forming a first shunt tap of the inductor in the first RDL, the first shunt tap coupled to the first terminal and the second terminal. The method also comprises coupling the die to the package substrate. The method also comprises coupling the first switch to the first shunt tap.

Description

BRIEF DESCRIPTION OF THE FIGURES
(1) FIG. **1** is a schematic diagram of an exemplary impedance tuner circuit that includes switchable inductors that are programmable to change frequency response of the impedance tuner circuit;
(2) FIGS. **2**A and **2**B are graphs illustrating inductance and quality (Q) factor as a function of frequency of an inductive path of the impedance tuner circuit in FIG. **1**;
(3) FIGS. **3**A and **3**B are side views of an integrated circuit (IC) package that includes an impedance tuner circuit that includes a tunable inductor formed in a redistribution layer (RDL) in a package substrate, wherein the tunable inductor includes a plurality of complementary conductive sections that each include ends adjacent to each other to form complementary shunt taps configured to be shunted by respective switches of the impedance tuner circuit in a semiconductor die ("die") coupled to the package substrate;
(4) FIG. **4** is a circuit diagram of an exemplary impedance tuner circuit that can be provided in the IC package in FIGS. **3**A and **3**B;
(5) FIGS. **5**A and **5**B are graphs illustrating inductance and Q factor as a function of frequency of an inductive path of the impedance tuner circuit in the IC package in FIGS. **3**A and **3**B;
(6) FIG. **6**A is a perspective view of another exemplary two-dimensional (2D) tunable inductor that can be formed in a RDL of an IC package;
(7) FIG. **6**B is a side view of another exemplary IC package that includes an impedance tuner circuit that includes the tunable inductor in FIG. **6**A;

(8) FIG. **7**A is a perspective view of another exemplary three-dimensional (3D) tunable inductor that can be formed in multiple RDLs of an IC package;
(9) FIG. **7**B is a side view of another exemplary IC package that includes an impedance tuner circuit that includes the tunable inductor in FIG. **7**A;
(10) FIG. **8**A is a perspective view of another exemplary 3D tunable inductor that can be formed in multiple RDLs of an IC package;
(11) FIG. **8**B is a side view of another exemplary IC package that includes an impedance tuner circuit that includes the tunable inductor in FIG. **8**A;
(12) FIG. **9** is a flowchart illustrating an exemplary process of fabricating an IC package that includes an impedance tuner circuit that includes a tunable inductor formed in a RDL of a package substrate, wherein the tunable inductor includes one or more shunt taps configured to be shunted by respective switches of the impedance tuner circuit in a die coupled to the package substrate;
(13) FIGS. **10**A-**10**D is a flowchart illustrating another exemplary fabrication process of fabricating an IC package that includes an impedance tuner circuit that includes a tunable inductor formed in a RDL of a package substrate, wherein the tunable inductor includes one or more shunt taps configured to be shunted by respective switches of the impedance tuner circuit in a die coupled to the package substrate, including but not limited to the IC packages in FIGS. **3**A-**4** and **6**A-**8**B;
(14) FIGS. **11**A-**11**G illustrate exemplary fabrication stages according to the exemplary IC package fabrication process in FIGS. **10**A-**10**D; and
(15) FIG. **12** is a block diagram of an exemplary wireless communications device that can include an IC package that includes an impedance tuner circuit that includes a tunable inductor that is formed in a RDL of a package substrate of the IC package, including, but not limited to, the IC packages FIGS. **3**A-**4**, **6**A-**8**B, and **11**A-**11**G.

DETAILED DESCRIPTION

(16) With reference now to the drawing figures, several exemplary aspects of the present disclosure are described. The word "exemplary" is used herein to mean "serving as an example, instance, or illustration." Any aspect described herein as "exemplary" is not necessarily to be construed as preferred or advantageous over other aspects.
(17) Aspects disclosed herein include an integrated circuit (IC) package employing an on-package tunable inductor formed in a redistribution layer (RDL) for an impedance tuner circuit. Related methods are also disclosed. The IC package includes an impedance tuner circuit that includes a tunable inductor that can be tuned to change the frequency response of the impedance tuner circuit. To reduce circuit area needed to provide the tunable inductor, the tunable inductor is integrated in an IC package. In exemplary aspects, the tunable inductor is formed in a RDL of a package substrate of the IC package. The IC package also includes a semiconductor die ("die") that includes other components of the impedance tuner circuit coupled to the tunable inductor by the die being coupled to the package substrate. In this manner, by the tunable inductor being formed in a RDL in the package substrate, the signal path lengths between the tunable inductor and other components of the tunable impedance circuit are reduced, thereby reducing the inductance path resistance and improving the quality (Q) factor of the tunable inductor. Also, by forming the tunable inductor in a RDL of the IC package, the tunable inductor can be formed of the desired geometry to provide the desired inductance frequency response.
(18) In exemplary aspects, the tunable inductor can be formed in the RDL of a package substrate of an IC package as a metal line having a first terminal and second terminal that can be coupled to a circuit to couple the inductance of the tunable inductor to the circuit. The tunable inductor is also formed in the RDL to have one or more shunt taps. A shunt tap is two metal lines formed in the tunable inductor that have ends adjacent to each other, but forming a short circuit. One or more switches in an active semiconductor layer in a front-end-of-line (FEOL) of the die as part of the impedance tuner circuit are coupled to respective shunt taps in the tunable inductor through signal path routing in the package substrate between the die and the tunable inductor in the RDL. The

switches in the die can be programmed/controlled to either shunt the shunt tap or leave their respective, coupled shunt tap as an open circuit, to change the inductance of the inductor and shift the frequency response of the impedance tuner circuit. In this manner, the switches allow the inductance of the inductor to be tuned through the controlling of the shunting or not shunting of the one or more shunt taps in the inductor. The tunable inductor being disposed in the RDL in the package substrate of the IC package locates the shunt taps closer to respective switches in the die to reduce inductance path length, thereby reducing resistance and improving inductance Q factor. Also, because the inductor is formed as a metal line in the RDL, the inductor and its shunt taps can be formed in the RDL of the desired geometry and connectivity arrangement as part of fabricating the package substrate, to provide the desired inductance tunability of the inductor.

(19) Before discussing examples of IC packages employing on-package tunable inductors formed in a redistribution layer (RDL) for an impedance tuner circuit starting at FIG. **3**A, an exemplary impedance tuner circuit and its performance in FIGS. **1**-**2**B is first described.

(20) In this regard, FIG. **1** is a schematic diagram of an exemplary impedance tuner circuit **100** that can be employed in a RF transceiver circuit. The impedance tuner circuit **100** includes switchable inductors L.sub.1-L.sub.4 that are each individually selectable through respective switches SW.sub.1-SW.sub.4 included in the impedance tuner circuit **100** to change frequency response of the impedance tuner circuit **100**. As shown in FIG. **1**, the impedance tuner circuit **100** includes an input node **102** configured to receive a radio-frequency (RF) signal **104** and an output node **106**. The impedance tuner circuit **100** filters the RF signal **104** received on the input node **102** into a filtered RF signal **104**F on the output node **106** depending on the selected arrangement of involvement of passive filtering components, including the switchable inductors L.sub.1-L.sub.4. In this regard, the impedance tuner circuit **100** has a signal path **108** between the input node **102** and the output node **106** that includes a first, variable capacitor C.sub.1 connected in series. A second, fixed capacitor C.sub.2 is connected in parallel to the first, variable capacitor C.sub.1. A switch S.sub.5 is connected in parallel to the variable capacitor C.sub.1, to the input node **102** and the output node **106**. When switch S.sub.5 is closed, the input node **102** is shorted to the output node **106**, thereby allowing the RF signal **104** to bypass the variable capacitor C.sub.1 and fixed capacitor C.sub.2. When switch S.sub.5 is open, the variable capacitor C.sub.1 and fixed capacitor C.sub.2 are present in the signal path **108** contributing to the filtering of the RF signal **104** to the filtered RF signal **104**F.

(21) With continuing reference to FIG. **1**, the impedance tuner circuit **100** includes switches S.sub.1-S.sub.4 that are connected in series to the inductors L.sub.1-L.sub.4, between the inductors L.sub.1-L.sub.4 and the respective input node **102** and output node **106**. In this regard, when any of the switches S.sub.1-S.sub.4 are closed, their respective coupled inductors L.sub.1-L.sub.4 are connected in parallel to the respective input node **102** and output node **106**, thus affecting the filtering of the RF signal **104** into the filtered RF signal **104**F. When any of the switches S.sub.1-S.sub.4 are open, their respective coupled inductors L.sub.1-L.sub.4 are disconnected to the respective input node **102** and output node **106**, thus not affecting the filtering of the RF signal **104** into the filtered RF signal **104**F. Thus, the frequency response of the impedance tuner circuit **100** is programmable based on the ability to selectively control the opening or closing of the switches S.sub.1-S.sub.5.

(22) FIGS. **2**A and **2**B are graphs **200**, **202** illustrating inductance and quality (Q) factor, respectively, as a function of frequency of an inductive path of the inductor L.sub.1 in the impedance tuner circuit **100** in FIG. **1**. Curve **204** in the graph **200** in FIG. **2**A illustrates the inductance (L) on the input node **102** of the impedance tuner circuit **100** from inductor L.sub.1 as a function of frequency (f) if switch SW.sub.1 were not present and the inductor L.sub.1 was directly coupled to the input node **102**. Curve **206** in graph **202** in FIG. **2**B illustrates the increase in the inductance on the input node **102** as a function of the presence of switch SW.sub.1 closed coupling inductor L.sub.1 to the input node **102** through switch SW.sub.1. Curve **208** in the graph **202** in

FIG. **2**B illustrates the quality (Q) factor of inductance (L) on the input node **102** of the impedance tuner circuit **100** from inductor L.sub.1 as a function of frequency (f) if switch SW were not present and the inductor L.sub.1 was directly coupled to the input node **102**. Curve **208** in the graph **202** in FIG. **2**B illustrates the quality (Q) factor of inductance (L) on the input node **102** of the impedance tuner circuit **100** from inductor L.sub.1 as a function of frequency (f) if switch SW.sub.1 were not present and the inductor L.sub.1 was directly coupled to the input node **102**. Curve **210** illustrates the Q factor of inductance (L) on the input node **102** as a function of the presence of switch SW.sub.1 being closed and coupling inductor L.sub.1 to the input node **102** through switch SW.sub.1. Thus, as shown in curve **210** FIG. **2**B, the Q factor of impedance on the input node **102** from inductor L.sub.1 decreases by the presence of switch SW.sub.1 as compared to the Q factor of impedance on the input node **102** from inductor L.sub.1 shown in curve **208** when the inductor L.sub.1 is directly coupled to the input node **102**.
(23) Thus, switch SW.sub.1 in the impedance tuner circuit **100** in FIG. **1** provides the capability of selectively coupling or not coupling the inductance from inductor L.sub.1 to the input node **102** for tuning the frequency response of the impedance tuner circuit **100**. However, the presence of switch SW.sub.1 increases the path length between the input node **102** and the inductor L.sub.1 that results in reduction of Q factor of inductor L.sub.1 on the input node **102** due to the increase in resistance in the inductance path to inductor L.sub.1. The inductors L.sub.1-L.sub.4 in the impedance tuner circuit **100** in FIG. **1** are located external to an IC package **112** that includes the switches SW.sub.2-SW.sub.4 and other components of the impedance tuner circuit **100** in FIG. **1** due to the size of the inductors L.sub.1-L.sub.4. Also, the losses in switch SW.sub.1 results in reduction of Q factor of inductor L.sub.1 on the input node **102**. Losses in Q factor of inductors L.sub.2-L.sub.4 on the respective input node **102** and output node **106** also occur due to the presence of respective switches SW.sub.2-SW.sub.4 in the impedance tuner circuit **100** in FIG. **1**. It is desired to reduce these losses to improve the Q factor of inductance in an impedance tuner circuit, such as the impedance tuner circuit **100** in FIG. **1**, without losing the capability of selecting tuning inductance to tune the frequency response of the impedance tuner circuit **100**.
(24) In this regard, FIGS. **3**A and **3**B are side views of an IC package **300** that includes an impedance tuner circuit **302** that includes a tunable inductor **304** ("inductor **304**") formed in a RDL **306** of a package substrate **308**. FIG. **3**A is a functional diagram of the IC package **300**. FIG. **3**B is a schematic diagram of a side view of IC package **300** from the cross-section line A.sub.1-A.sub.1′ in FIG. **3**A. As discussed in more detail below, the inductor **304** can be tuned to change the inductance and frequency response of the impedance tuner circuit **302**. The inductor **304** is integrated in the IC package **300**, and more particularly the RDL **306** of the package substrate **308** of the IC package **300**. An RDL can be an extra metallization layer that is formed in a package substrate that includes formed metal lines for redistributing metal pads for making external connections to other locations in a package substrate. The fabrication technology used for forming metal line in a RDL can also be used to form the inductor **304** in the RDL **306** of the package substrate **308** of the IC package **300**. As shown in FIG. **3**B, by the inductor **304** being formed in the RDL **306** of the package substrate **308** of the IC package **300**, the impedance tuner circuit **302** can be completely provided in the IC package **300**. The IC package **300** can be coupled to a printed circuit board (PCB) **310** through external interconnects **312** (e.g., solder balls) coupled to the package substrate **308** to couple the impedance tuner circuit **302** to other circuits coupled to the PCB **310**.
(25) With continuing reference to FIGS. **3**A and **3**B, as also discussed in more detail below, the inductor **304** in the RDL **306** of the package substrate **308** is coupled to switches **314**(**1**), **314**(**2**) formed in a die **316** of the IC package **300**. FIG. **3**B only shows switch **314**(**1**) among the switches **314**(**1**), **314**(**2**) shown in FIG. **3**A, according to the side view along the cross-section line A.sub.1-A.sub.1′ in FIG. **3**B. The opening and closing of the switches **314**(**1**), **314**(**2**) are controllable to selectively shunt or not shunt respective shunt taps **318**(**1**), **318**(**2**) formed in the inductor **304** to

shift the inductance frequency response of the impedance tuner circuit **302**. In this example, two shunt taps **318**(**1**), **318**(**2**) are formed in the inductor **304**, but note that one (1) or more than two (2) shunt taps could be formed in the inductor **304**. The number of shunt taps formed in the inductor **304** affects the number of programmable settings for shifting the inductance frequency response inductance of the inductor **304**. In this manner, by the inductor **304** being formed in the RDL **306** of the package substrate **308** of the IC package **300**, the signal path lengths between the inductor **304** and the respective switches **314**(**1**), **314**(**2**) are reduced, thereby reducing the inductance path resistance of the shunt taps **318**(**1**), **318**(**2**) and improving the Q factor of the inductor **304**. Also, as discussed in more detail below, by forming the inductor **304** in the RDL **306** of the IC package **300**, the inductor **304** can be formed of the desired geometry to provide the desired inductance frequency response. In this example, because the inductor **304** is formed in a single RDL **306** of the IC package **300**, the inductor **304** is a two-dimensional (2D) inductor **304**.
(26) With continuing reference to FIGS. **3**A and **3**B, in this example, the inductor **304** formed in the RDL **306** of the package substrate **308** of the IC package **300** is formed as a metal line **320** having a first terminal **322**(**1**) and second terminal **322**(**2**) that are coupled to the impedance tuner circuit **100**. The shunt taps **318**(**1**), **318**(**2**) are both coupled in parallel to the first and second terminals **322**(**1**), **322**(**2**). The shunt taps **318**(**1**), **318**(**2**) are each formed of two (2) respective shunt metal lines **324**(**1**)-**324**(**2**), **326**(**1**)-**326**(**2**) formed in the inductor **304** and that have first and second ends **328**(**1**)-**328**(**2**) and first and second ends **330**(**1**)-**330**(**2**) adjacent to each other, but forming a short circuit. As shown in FIG. **3**A, the switches **314**(**1**), **314**(**2**) in an active semiconductor layer **332** in a front-end-of-line (FEOL) **334** of the die **316** as part of the impedance tuner circuit **302** are coupled to respective ends **328**(**1**)-**328**(**2**), **330**(**1**)-**330**(**2**) of the respective shunt taps **318**(**1**), **318**(**2**) in the inductor **304**. The switches **314**(**1**), **314**(**2**) each include respective first and second switch terminals **331**(**1**)-**331**(**2**), **333**(**1**), **333**(**2**) that are coupled to respective first and second ends **328**(**1**)-**328**(**2**) and first and second ends **330**(**1**)-**330**(**2**) of the respective shunt taps **318**(**1**)-**318**(**2**). As shown in FIGS. **3**A and **3**B, the switches **314**(**1**), **314**(**2**) are coupled to the respective ends **328**(**1**)-**328**(**2**), **330**(**1**)-**330**(**2**) of the respective shunt taps **318**(**1**), **318**(**2**) through signal path routing in the package substrate **308** between the die **316** and the inductor **304** in the RDL **306**. In this example, as shown in FIG. **3**B, the package substrate **308** includes a metallization layer **336** that is adjacent to the RDL **306** and disposed between the RDL **306** and the die **316** in a vertical direction (Z-axis direction). The metallization layer **336** includes metal interconnects that are located behind the metal interconnects **338**(**1**), **338**(**2**) in the Y-axis direction that are coupled to the respective ends **328**(**1**)-**328**(**2**), **330**(**1**)-**330**(**2**) of the respective shunt taps **318**(**1**), **318**(**2**) and to die interconnects **340** of the die **316** that are coupled to respective switches **314**(**1**), **314**(**2**), to couple the shunt taps **318**(**1**), **318**(**2**) of the inductor **304** to the switches **314**(**1**), **314**(**2**).
(27) In this manner, as shown in FIGS. **3**A and **3**B, the switches **314**(**1**), **314**(**2**) in the die **316** can be programmed/controlled to either shunt the shunt taps **318**(**1**), **318**(**2**) or leave their respective, coupled shunt taps **318**(**1**), **318**(**2**) as an open circuit, or a mixture thereof, to shift the inductance of the inductor **304** and its inductance frequency response in the impedance tuner circuit **302**. In this manner, the switches **314**(**1**), **314**(**2**) allow the inductance of the inductor **304** to be tuned through the controlling of the shunting or not shunting of the one or more shunt taps **318**(**1**), **318**(**2**) in the inductor **304**. The inductor **304** being disposed in the RDL **306** in the package substrate **308** of the IC package **300** locates the shunt taps **318**(**1**), **318**(**2**) closer to respective switches **314**(**1**), **314**(**2**) in the die **316** to reduce inductance path length, thereby reducing resistance and improving inductance Q factor of the inductor **304**. Also, because the inductor **304** is formed as a metal line **320** in the RDL **306**, the inductor **304** and its shunt taps **318**(*l*), **318**(**2**) can be formed in the RDL **306** of the desired geometry and connectivity arrangement as part of fabricating the package substrate **308**, to provide the desired inductance tunability of the inductor **304**.
(28) Note that metal lines **342**(**1**), **324**(**2**) in the inductor **304** could be classified as a shunt tap wherein a switch coupled be coupled to the metal lines **342**(**1**), **342**(**2**) to control the shunting of the

metal lines **342**(**1**), **342**(**2**). The shunting of the inductor **304** can be controlled depending on the circuit in which the inductor **304** is included, to control inductance while having a two (2) terminal connection to the circuit. Note that as shown in FIG. **3**A, the first and second terminals **322**(**1**), **322**(**2**) are formed in respective metal lines **342**(**1**), **342**(**2**) that extend along longitudinal axis L.sub.1 in the X-axis direction. The shunt taps **318**(**1**), **318**(**2**) each extend along longitudinal axes L.sub.2, L.sub.3 that are parallel to the longitudinal axis L.sub.1. However, as discussed in other examples of tunable inductors below, shunt taps in a tunable inductor may extend in non-parallel axes to the metal lines in which the first and second terminals of the tunable inductor are formed.
(29) FIG. **4** is a circuit diagram of the impedance tuner circuit **302** that includes the inductor **304** in the IC package **300** in FIGS. **3**A and **3**B. As shown in FIG. **4**, the impedance tuner circuit **302** includes an input node **400** configured to receive a RF signal **402** and an output node **404** configured to provide a filtered RF signal **402**F. The die **316** includes switches **314**(**1**), **314**(**2**) as part of the impedance tuner circuit **302** that are coupled to the respective shunt taps **318**(**1**), **318**(**2**) of the tunable inductor **304** in the package substrate **308**. An optional external inductor L.sub.5 is coupled to the first terminal **322**(**1**). An optional resistor R.sub.1 is coupled to the second terminal **322**(**2**) of the inductor **304**. Switch **314**(**3**) is coupled to the shunt tap **318**(**1**) and a first node **406**(**1**). Switch **314**(**4**) is coupled to the shunt tap **318**(**2**) and a second node **406**(**2**). The switches **314**(**1**), **314**(**2**) can be controlled to be open or closed to either not shunt or shunt the respective shunt taps **318**(**1**), **318**(**2**) in the inductor **304** to shift the inductance frequency response of the inductor **304**. The impedance tuner circuit **302** also includes a variable capacitor C.sub.3 that is configured to be selectively coupled in series or bypass between the input node **400** and the output node **404**.
(30) FIGS. **5**A and **5**B are graphs **500**, **502** illustrating inductance and Q factor, respectively, as a function of frequency of an inductive path of the inductor **304** in the impedance tuner circuit **302** in FIGS. **3**A-**4**. Curves **504**(**1**)-**504**(**4**) in the graph **500** in FIG. **5**A illustrates the inductance (L) of the impedance tuner circuit **302** in FIG. **4** as a function of frequency based on ('1') and off ('0') combinations of switches **314**(**1**), **314**(**2**) coupled to the shunt taps **318**(**1**), **318**(**2**) as shown. As shown in FIG. **5**A, the controlling of the switches **314**(**1**), **314**(**2**) in different on/off states shifts the inductance (L) of the impedance tuner circuit **302**. Curve **504**(**5**) in the graph **500** in FIG. **5**A illustrates the inductance (L) on the input node **400** of the impedance tuner circuit **302** in FIG. **4** as a function of frequency based only inductor L.sub.5 without the presence of inductor **304**.
(31) Curves **506**(**1**)-**506**(**4**) in graph **502** in FIG. **5**B illustrate the Q factor of inductance (L) of the impedance tuner circuit **100** from inductor **304** as a function of frequency (f) based on the same on ('1') and off ('0') combinations of switches **314**(**1**), **314**(**2**) that are related to respective inductance curves **504**(**1**)-**504**(**4**) in FIG. **2**A. As shown in FIG. **5**B, the controlling of the switches **314**(**1**), **314**(**2**) in different on/off states shifts the inductance (L) of the impedance tuner circuit **302**. Curve **506**(**5**) in the graph **502** in FIG. **5**B illustrates the Q factor of inductance (L) of the impedance tuner circuit **302** in FIG. **4** as a function of frequency based only inductor L.sub.5 without the presence of inductor **304**. Note that the Q factor is higher in curves **506**(**1**)-**506**(**4**) than curve **506**(**5**).
(32) FIG. **6**A is a perspective view of another exemplary 2D tunable inductor **604** ("inductor **604**") that can be formed in a RDL of an IC package. FIG. **6**B is a side view of another exemplary IC package **600** that includes an impedance tuner circuit **602** similar to the impedance tuner circuit **302** in the IC package **300** in FIG. **3**B. The IC package **600** in FIG. **6**B includes a RDL **606** that includes the inductor **604** in FIG. **6**A. FIG. **6**B is a schematic diagram of a side view IC package **600** from the cross-section line A.sub.2-A.sub.2′ in the inductor **604** in FIG. **6**A. Common elements between the IC package **300** in FIG. **3**B and the IC package **600** in FIG. **6**B are shown with common element numbers and not re-described. Note that as shown in FIG. **6**A, like the inductor **304** in FIGS. **3**A and **3**B, the first and second terminals **322**(**1**), **322**(**2**) of the inductor **604** are formed in respective metal lines **342**(**1**), **342**(**2**) that extend along the longitudinal axis L.sub.1 in the X-axis direction. However, as also shown in FIG. **6**A, in this example of inductor **604**, its shunt

taps **618**(**1**), **618**(**2**) each extend along longitudinal axes longitudinal axes L.sub.4, L.sub.5 that are orthogonal to the longitudinal axis L.sub.1 as opposed to parallel like in the inductor **304** in FIGS. **3**A and **3**B.

(33) With reference to FIGS. **6**A and **6**B, the inductor **604** in the RDL **606** of the package substrate **608** is coupled to switches **314**(**1**), **314**(**2**) formed in the die **316** of the IC package **600**. FIG. **6**B only shows a diagram of switch **314**(**1**) among the switches **314**(**1**), **314**(**2**) shown in FIG. **3**A, according to the side view along the cross-section line A.sub.2-A.sub.2′ in the inductor **604** in FIG. **6**B. The opening and closing of the switches **314**(**1**), **314**(**2**) are controllable to selectively shunt or not shunt respective shunt taps **618**(**1**), **618**(**2**) formed in the inductor **604** to shift the inductance frequency response of the impedance tuner circuit **302**.

(34) As shown in FIG. **6**A, the inductor **604** formed in the RDL **606** of the package substrate **608** of the IC package **600** is formed as a metal line **620** having the first terminal **322**(**1**) and second terminal **322**(**2**) that is coupled to the impedance tuner circuit **602**. The shunt taps **618**(**1**), **618**(**2**) are both coupled in series to the first and second terminals **322**(**1**), **322**(**2**) in this example. The shunt taps **618**(**1**), **618**(**2**) are each formed of two (2) respective shunt metal lines **624**(**1**)-**624**(**2**), **626**(**1**)-**626**(**2**) formed in the inductor **604** and have first and second ends **628**(**1**)-**628**(**2**) and first and second ends **630**(**1**)-**630**(**2**) adjacent to each other, but forming a short circuit. The switches **314**(**1**), **314**(**2**) are coupled to the respective ends **628**(**1**)-**628**(**2**), **630**(**1**)-**630**(**2**) of the respective shunt taps **618**(**1**), **618**(**2**) through signal path routing in the package substrate **608** between the die **316** and the inductor **604** in the RDL **606**. The switches **314**(**1**), **314**(**2**) in the die **316** can be programmed/controlled to either shunt the shunt taps **618**(**1**), **618**(**2**) or leave their respective, coupled shunt taps **618**(**1**), **618**(**2**) as an open circuit, or a mixture thereof, to shift the inductance of the inductor **604** and its inductance frequency response in an impedance tuner circuit **602**.

(35) Note that like the inductor **304** in FIGS. **3**A and **3**B, the metal lines **342**(*l*), **324**(**2**) in the inductor **604** could be classified as a shunt tap wherein a switch coupled be coupled to the metal lines **342**(**1**), **342**(**2**) to control the shunting of the metal lines **342**(**1**), **342**(**2**). The shunting of the inductor **604** can be controlled depending on the circuit in which the inductor **604** is included, to control inductance while having a two (2) terminal connection to the circuit.

(36) FIG. **7**A is a perspective view of an exemplary 3D tunable inductor **704** ("inductor **704**") that can be formed in multiple RDLs of an IC package. FIG. **7**B is a side view of another exemplary IC package **700** that includes an impedance tuner circuit **702** similar to the impedance tuner circuit **302** in the IC package **300** in FIG. **3**B. The IC package **700** in FIG. **7**B includes RDL **306** and an additional RDL **706** that includes the inductor **704** in FIG. **7**A. FIG. **7**B is a schematic diagram of a side view IC package **700** from the cross-section line A.sub.3-A.sub.3′ in the inductor **704** in FIG. **7**A. Common elements between the IC package **700** in FIG. **7**B and the IC package **300** in FIG. **3**B are shown with common element numbers and not re-described. The inductor **704** in FIG. **7**A is similar to the inductor **304** in FIGS. **3**A and **3**B. However, as discussed below and shown in FIGS. **7**A and **7**B, the inductor **704** also includes an additional shunt tap **718** that is disposed in a second RDL **706** adjacent to the first RDL **306** in the package substrate **708** of the IC package **700**. Also, as also shown in FIG. **7**A, in this example of inductor **704**, the shunt tap **718** disposed in the second RDL **706** extend along longitudinal axis L.sub.6 that intersects the longitudinal axis L.sub.1 of the first and second terminals **322**(**1**), **322**(**2**).

(37) With reference to FIGS. **7**A and **7**B, the inductor **704** in the RDLs **306**, **706** of the package substrate **708** is coupled to switches **314**(**1**), **314**(**2**) formed in the die **316** of the IC package **700**. FIG. **7**B only shows switch **314**(**1**) among the switches **314**(**1**), **314**(**2**), according to the side view along the cross-section line A.sub.3-A.sub.3′ in the inductor **704** in FIG. **7**B. An additional switch **714** disposed in the die **316** is coupled to third shunt tap **718** of the inductor **704**. The opening and closing of the switches **314**(**1**), **314**(**2**), **714** are controllable to selectively shunt or not shunt respective shunt taps **318**(**1**), **318**(**2**), **718** formed in the inductor **704** to shift the inductance frequency response of the impedance tuner circuit **702**.

(38) As shown in FIG. **7**A, the inductor **704** formed in the RDLs **306**, **706** of the package substrate **708** of the IC package **700** is formed as respective metal line **320** having the first terminal **322**(**1**) and second terminal **322**(**2**) and metal line **720** that is coupled to the impedance tuner circuit **702**. The shunt tap **718** is coupled in parallel to the first and second terminals **322**(**1**), **322**(**2**) of the inductor **704** in this example. The shunt tap **718** is formed of two (2) shunt metal lines **724**(**1**), **724**(**2**) formed in the inductor **704** and that have first and second ends **728**(**1**), **728**(**2**) adjacent to each other, but forming a short circuit. Vias **730**(**1**), **730**(**2**) couple the respective first and second ends **728**(**1**), **728**(**2**) of the shunt tap **718** to the respective first and second terminals **322**(**1**), **322**(**2**). The switch **714** is coupled to the respective ends **728**(**1**), **728**(**2**) of the shunt tap **718** through signal path routing in the package substrate **708** between the die **316** and the inductor **704** in the first and second RDLs **306**, **706**. The switches **314**(**1**), **314**(**2**), **714** in the die **316** can be programmed/controlled to either shunt the shunt taps **318**(**1**), **318**(**2**), **718** or leave their respective, coupled shunt taps **318**(**1**), **318**(**2**), **718** as an open circuit, or a mixture thereof, to shift the inductance of the inductor **704** and its inductance frequency response in an impedance tuner circuit **702**.
(39) Note that metal lines **342**(**1**), **324**(**2**) in the inductor **704** could be classified as a shunt tap wherein a switch coupled be coupled to the metal lines **342**(**1**), **342**(**2**) to control the shunting of the metal lines **342**(**1**), **342**(**2**). The shunting of the inductor **704** can be controlled depending on the circuit in which the inductor **704** is included, to control inductance while having a two (2) terminal connection to the circuit.
(40) FIG. **8**A is a perspective view of another exemplary 3D tunable inductor **804** ("inductor **804**") that can be formed in multiple RDLs of an IC package. FIG. **8**B is a side view of another exemplary IC package **800** that includes an impedance tuner circuit **802** similar to the impedance tuner circuit **302** in the IC package **700** in FIG. **7**B. The IC package **800** in FIG. **8**B includes RDL **306**, and an additional RDL **806** that includes the inductor **804** in FIG. **8**A. FIG. **8**B is a schematic diagram of a side view IC package **800** from the cross-section line A.sub.4-A.sub.4′ in the inductor **804** in FIG. **8**A. Common elements between the IC package **800** in FIG. **8**B and the IC packages **300**, **700** in FIGS. **3**B and **7**B are shown with common element numbers and not re-described. The inductor **804** in FIG. **8**A is similar to the inductors **304**, **704** in FIGS. **3**B and **7**B. However, as discussed below and shown in FIGS. **8**A and **8**B, the inductor **804** also includes two additional shunt taps **818**(**1**), **818**(**2**) that are both disposed in the second RDL **806** adjacent to the first RDL **306** in the package substrate **808** of the IC package **800**. Also, as also shown in FIG. **8**A, in this example of inductor **804**, the shunt taps **818**(**1**), **818**(**2**) disposed in the second RDL **806** extend along longitudinal axes longitudinal axis L.sub.7, L.sub.8 that are parallel to longitudinal axis L.sub.7, L.sub.8 of the shunt taps **318**(**1**), **318**(**2**).
(41) With reference to FIGS. **8**A and **8**B, the inductor **804** in the RDLs **306**, **806** of the package substrate **808** is coupled to switches **314**(**1**), **314**(**2**), **814**(**1**), **814**(**2**) formed in the die **316** of the IC package **800**. FIG. **7**B only shows switch **314**(**1**) among the switches **314**(**1**), **314**(**2**), **814**(**1**), **814**(**2**) according to the side view along the cross-section line A.sub.4-A.sub.4′ in the inductor **804** in FIG. **8**B. Switches **814**(**1**), **814**(**2**) disposed in the die **316** are coupled to third and fourth shunt taps **818**(**1**), **818**(**2**) of the inductor **804** in the second RDL **806**. The opening and closing of the switches **314**(**1**), **314**(**2**), **814**(**1**), **814**(**2**) are controllable to selectively shunt or not shunt respective shunt taps **318**(**1**), **318**(**2**), **818**(**1**), **818**(**2**) formed in the inductor **804** to shift the inductance frequency response of the impedance tuner circuit **802**.
(42) As shown in FIG. **8**A, the inductor **804** formed in the RDLs **306**, **806** of the package substrate **808** of the IC package **800** is formed as the metal line **320** having the first terminal **322**(**1**) and second terminal **322**(**2**) and a metal line **820** that is coupled to the impedance tuner circuit **802**. The shunt taps **818**(**1**), **818**(**2**) are coupled in parallel to the first and second terminals **322**(**1**), **322**(**2**) of the inductor **804** in this example. The shunt taps **818**(**1**), **818**(**2**) are formed of two (2) respective shunt metal lines **824**(**1**)-**824**(**2**), **826**(**1**)-**826**(**2**) formed in the inductor **804** and that have first and

second ends **828**(**1**)-**828**(**2**), **830**(**1**)-**830**(**2**) adjacent to each other, but forming a short circuit. A via **830** couples the respective first terminals **322**(**1**) to the metal line **820** in the second RDL **806**. The switches **814**(**1**), **814**(**2**) are coupled to the respective ends **828**(**1**)-**828**(**2**), **830**(**1**)-**830**(**2**) of the respective shunt taps **818**(**1**), **818**(**2**) through signal path routing in the package substrate **808** between the die **316** and the inductor **804** in the first and second RDLs **306**, **806**. The switches **314**(**1**), **314**(**2**), **814**(**1**), **814**(**2**) in the die **316** can be programmed/controlled to either shunt the shunt taps **318**(**1**), **318**(**2**), **818**(**1**), **818**(**2**) or leave their respective, coupled shunt taps **318**(**1**), **318**(**2**), **818**(**1**), **818**(**2**) as an open circuit, or a mixture thereof, to shift the inductance of the inductor **804** and its inductance frequency response in an impedance tuner circuit **802**.

(43) Note that metal lines **342**(**1**), **324**(**2**) in the inductor **804** could be classified as a shunt tap wherein a switch coupled be coupled to the metal lines **342**(**1**), **342**(**2**) to control the shunting of the metal lines **342**(**1**), **342**(**2**). The shunting of the inductor **804** can be controlled depending on the circuit in which the inductor **804** is included, to control inductance while having a two (2) terminal connection to the circuit.

(44) FIG. **9** is a flowchart illustrating an exemplary fabrication process **900** of fabricating an IC package that includes an impedance tuner circuit that includes the tunable formed in a RDL of a package substrate. The tunable inductor includes one or more shunt taps configured to be shunted by respective switches disposed in a die coupled to the package substrate. The process **900** is discussed below with regard to the IC package **300** in FIGS. **3**A and **3**B, but note that the process **900** is not limited to the IC package **300** in FIGS. **3**A and **3**B. The process **900** in FIG. **9** can be employed to fabricate the IC packages **600**, **700**, and **800** in FIGS. **6**A-**8**B, respectively.

(45) In this regard, as shown in FIG. **9**, a first step of the fabrication process **900** is forming a first switch **314**(**1**) in a die **316** (block **902** in FIG. **9**). A next step in the fabrication process **900** is forming an inductor **304** in a package substrate **308** (block **904** in FIG. **9**). Forming the inductor **304** can include the steps of forming a first metal line **320** in the RDL **306** of the package substrate **308**, the first metal line **320** comprising a first terminal **322**(**1**) and a second terminal **322**(**2**) (block **906** in FIG. **9**), and forming a first shunt tap **318**(**1**) in the RDL **306**, the first shunt tap **318**(**1**) coupled to the first terminal **322**(**1**) and the second terminal **322**(**2**) (block **908** in FIG. **9**). A next step in the fabrication process **900** is coupling the die **316** to the package substrate **308** (block **910** in FIG. **9**). A next step in the fabrication process **900** is coupling the first switch **314**(**1**) to the first shunt tap **318**(**1**) (block **912** in FIG. **9**).

(46) An IC package that includes an impedance tuner circuit that includes the tunable formed in a RDL of a package substrate, including, but not limited, to the IC packages **300**, **600**, **700**, and **800** in FIGS. **3**A-**3**B, and **6**A-**8**B, can be fabricated in other fabrication processes. For example, FIGS. **10**A-**10**D is a flowchart illustrating an exemplary fabrication process **1000** of fabricating an IC package that includes an impedance tuner circuit that includes the tunable formed in a RDL of a package substrate. FIGS. **11**A-**11**G illustrate exemplary fabrication stages **1100**A-**1100**G according to the exemplary fabrication process **1000** in FIGS. **10**A-10D. The fabrication process **1000** in FIGS. **10**A-**10**D is discussed below with reference to the IC package **300** in FIGS. **3**A and **3**B, but the fabrication process **1000** is not limited to fabricating the IC package **300** in FIGS. **3**A and **3**B.

(47) In this regard, as shown in the exemplary fabrication stage **1100**A in FIG. **11**A, a first step of the fabrication process **1000** can be to create the die **316** and form the switches **314**(**1**), **314**(**2**) in the active semiconductor layer **332** (e.g., FEOL) of the die **316** (block **1002** in FIG. **10**A). Then, as shown in the exemplary fabrication stage **1100**B in FIG. **11**B, a next step of the fabrication process **1000** can be to form the metal interconnects **1102** in a metallization layer **1104** of the die **316** coupled to the switches **314**(**1**), **314**(**2**) and then form the die interconnects **340** for the die **316** (block **1004** in FIG. **10**A). Some of the die interconnects **340** are coupled to the metal interconnects **1102** that are coupled to the switches **314**(**1**), **314**(**2**) to provide a signal routing path between the switches **314**(**1**), **314**(**2**) and the shunt taps **318**(**1**), **318**(**2**) in the package substrate **308** in which the die **316** will be coupled.

(48) As shown in the exemplary fabrication stage **1100**C in FIG. **11**C, a next step of the fabrication process **1000** can be to form the inductor **304** in the RDL **306** of the package substrate **308** of the IC package **300** in FIG. **3**B (block **1006** in FIG. **10**A). The package substrate **308** is not shown in FIG. **11**C, but is shown in FIG. **3**B. The inductor is formed by forming the metal line **320** with the first and second terminals **322**(**1**), **322**(**2**). Then, as shown in the exemplary fabrication stage **1100**D in FIG. **11**D, a next step of the fabrication process **1000** can be to form the shunt taps **318**(**1**), **318**(**2**) in the inductor **304** (block **1008** in FIG. **10**B and block **1010** in FIG. **10**C). Only the first shunt tap **318**(**1**) is shown as being formed in the inductor **304** in the fabrication stage **1100**D in FIG. **11**D, but note that the second shunt tap **318**(**2**) can also be formed as shown in the fabrication stage **1100**E in FIG. **11**E. As shown in FIG. **11**D, the shunt tap **318**(**1**) is formed by forming shunt metal lines **324**(**1**), **324**(**2**) coupled to the respective first and second terminals **322**(**1**), **322**(**2**) with their ends **328**(**1**), **328**(**2**) adjacent to each other. The second shunt tap **318**(**2**) is similarly formed in the inductor **304** as shown in FIG. **11**E. Thus, when the die **316** is coupled to package substrate **308** that includes a RDL **306** with the inductor **304**, the die interconnects **340** that are coupled to the switches **314**(**1**), **314**(**2**) are coupled to the respective shunt taps **318**(**1**), **318**(**2**) of the inductor.
(49) Then, as shown in the exemplary fabrication stage **1100**F in FIG. **11**F, a next step of the fabrication process **1000** is to couple the die **316** to the package substrate **308** to couple the switches **314**(**1**), **314**(**2**) in the die **316** to the respective shunt taps **318**(**1**), **318**(**2**) in the RDL **306** of the package substrate **308** (block **1012** in FIG. **10**C). Then, as shown in the exemplary fabrication stage **1100**G in FIG. **11**G, a next step of the fabrication process **1000** is to form the external interconnects **312** coupled to the package substrate **308** and mounting of the IC package **300** to the PCB **310** (block **1014** in FIG. **10**D). Note that an external inductor **1106** may also be coupled to the PCB **310** and coupled to the impedance tuner circuit **302** in the IC package **300** through metal line routing through the PCB **310** and to an external interconnect **312**.
(50) The term “coupling” described herein can mean electrical coupling and/or physical coupling. Further, an IC package that includes an impedance tuner circuit that includes the tunable formed in a RDL of a package substrate, including but not limited to, the IC packages **300**, **600**, **700**, and **800** in FIGS. **3**A-**3**B and **6**A-**8**B, and according to the exemplary fabrication processes **900**, **1000** in FIGS. **9** and **10**A-**10**D, and according to any aspects disclosed herein, may be provided in or integrated into any electronic device. Examples, without limitation, include a set top box, an entertainment unit, a navigation device, a communications device, a fixed location data unit, a mobile location data unit, a global positioning system (GPS) device, a mobile phone, a cellular phone, a smart phone, a session initiation protocol (SIP) phone, a tablet, a phablet, a server, a computer, a portable computer, a mobile computing device, a wearable computing device (e.g., a smart watch, a health or fitness tracker, eyewear, etc.), a desktop computer, a personal digital assistant (PDA), a monitor, a computer monitor, a television, a tuner, a radio, a satellite radio, a music player, a digital music player, a portable music player, a digital video player, a video player, a digital video disc (DVD) player, a portable digital video player, an automobile, a vehicle component, avionics systems, a drone, and a multicopter.
(51) In this regard, FIG. **12** illustrates an exemplary wireless communications device **1200** as an electronic device that includes RF components formed from one or more integrated circuits (ICs) **1202**, wherein any of the ICs **1202** can be included in an IC package **1203** that includes an impedance tuner circuit that includes the tunable formed in a RDL of a package substrate, including but not limited to, the IC packages **300**, **600**, **700**, and **800** in FIGS. **3**A-**3**B and **6**A-**8**B. As shown in FIG. **12**, the wireless communications device **1200** includes a transceiver **1204** and a data processor **1206**. The data processor **1206** may include a memory to store data and program codes. The transceiver **1204** includes a transmitter **1208** and a receiver **1210** that support bi-directional communications. In general, the wireless communications device **1200** may include any number of transmitters **1208** and/or receivers **1210** for any number of communication systems and frequency bands. All or a portion of the transceiver **1204** may be implemented on one or more analog ICs,

RFICs, mixed-signal ICs, etc.

(52) The transmitter **1208** or the receiver **1210** may be implemented with a super-heterodyne architecture or a direct-conversion architecture. In the super-heterodyne architecture, a signal is frequency-converted between RF and baseband in multiple stages, e.g., from RF to an intermediate frequency (IF) in one stage, and then from IF to baseband in another stage. In the direct-conversion architecture, a signal is frequency-converted between RF and baseband in one stage. The super-heterodyne and direct-conversion architectures may use different circuit blocks and/or have different requirements. In the wireless communications device **1200** in FIG. **12**, the transmitter **1208** and the receiver **1210** are implemented with the direct-conversion architecture.

(53) In the transmit path, the data processor **1206** processes data to be transmitted and provides I and Q analog output signals to the transmitter **1208**. In the exemplary wireless communications device **1200**, the data processor **1206** includes digital-to-analog converters (DACs) **1212**(**1**), **1212**(**2**) for converting digital signals generated by the data processor **1206** into the I and Q analog output signals, e.g., I and Q output currents, for further processing.

(54) Within the transmitter **1208**, lowpass filters **1214**(**1**), **1214**(**2**) filter the I and Q analog output signals, respectively, to remove undesired signals caused by the prior digital-to-analog conversion. Amplifiers (AMPs) **1216**(**1**), **1216**(**2**) amplify the signals from the lowpass filters **1214**(**1**), **1214**(**2**), respectively, and provide I and Q baseband signals. An upconverter **1218** upconverts the I and Q baseband signals with I and Q transmit (TX) local oscillator (LO) signals from a TX LO signal generator **1222** through mixers **1220**(**1**), **1220**(**2**) to provide an upconverted signal **1224**. A filter **1226** filters the upconverted signal **1224** to remove undesired signals caused by the frequency upconversion as well as noise in a receive frequency band. A PA **1228** amplifies the upconverted signal **1224** from the filter **1226** to obtain the desired output power level and provides a transmit RF signal. The transmit RF signal is routed through a duplexer or switch **1230** and transmitted via an antenna **1232**.

(55) In the receive path, the antenna **1232** receives signals transmitted by base stations and provides a received RF signal, which is routed through the duplexer or switch **1230** and provided to a low noise amplifier (LNA) **1234**. The duplexer or switch **1230** is designed to operate with a specific receive (RX)-to-TX duplexer frequency separation, such that RX signals are isolated from TX signals. The received RF signal is amplified by the LNA **1234** and filtered by a filter **1236** to obtain a desired RF input signal. Downconversion mixers **1238**(**1**), **1238**(**2**) mix the output of the filter **1236** with I and Q RX LO signals (i.e., LO_I and LO_Q) from an RX LO signal generator **1240** to generate I and Q baseband signals. The I and Q baseband signals are amplified by AMPs **1242**(**1**), **1242**(**2**) and further filtered by lowpass filters **1244**(**1**), **1244**(**2**) to obtain I and Q analog input signals, which are provided to the data processor **1206**. In this example, the data processor **1206** includes analog-to-digital converters (ADCs) **1246**(**1**), **1246**(**2**) for converting the analog input signals into digital signals to be further processed by the data processor **1206**.

(56) In the wireless communications device **1200** of FIG. **12**, the TX LO signal generator **1222** generates the I and Q TX LO signals used for frequency upconversion, while the RX LO signal generator **1240** generates the I and Q RX LO signals used for frequency downconversion. Each LO signal is a periodic signal with a particular fundamental frequency. A TX phase-locked loop (PLL) circuit **1248** receives timing information from the data processor **1206** and generates a control signal used to adjust the frequency and/or phase of the TX LO signals from the TX LO signal generator **1222**. Similarly, an RX PLL circuit **1250** receives timing information from the data processor **1206** and generates a control signal used to adjust the frequency and/or phase of the RX LO signals from the RX LO signal generator **1240**.

(57) Those of skill in the art will further appreciate that the various illustrative logical blocks, modules, circuits, and algorithms described in connection with the aspects disclosed herein may be implemented as electronic hardware, instructions stored in memory or in another computer readable medium and executed by a processor or other processing device, or combinations of both.

The master and slave devices described herein may be employed in any circuit, hardware component, IC, or IC chip, as examples. Memory disclosed herein may be any type and size of memory and may be configured to store any type of information desired. To clearly illustrate this interchangeability, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their functionality. How such functionality is implemented depends upon the particular application, design choices, and/or design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the present disclosure.

(58) The various illustrative logical blocks, modules, and circuits described in connection with the aspects disclosed herein may be implemented or performed with a processor, a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Field Programmable Gate Array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A processor may be a microprocessor, but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration).

(59) The aspects disclosed herein may be embodied in hardware and in instructions that are stored in hardware, and may reside, for example, in Random Access Memory (RAM), flash memory, Read Only Memory (ROM), Electrically Programmable ROM (EPROM), Electrically Erasable Programmable ROM (EEPROM), registers, a hard disk, a removable disk, a CD-ROM, or any other form of computer readable medium known in the art. An exemplary storage medium is coupled to the processor such that the processor can read information from, and write information to, the storage medium. In the alternative, the storage medium may be integral to the processor. The processor and the storage medium may reside in an ASIC. The ASIC may reside in a remote station. In the alternative, the processor and the storage medium may reside as discrete components in a remote station, base station, or server.

(60) It is also noted that the operational steps described in any of the exemplary aspects herein are described to provide examples and discussion. The operations described may be performed in numerous different sequences other than the illustrated sequences. Furthermore, operations described in a single operational step may actually be performed in a number of different steps. Additionally, one or more operational steps discussed in the exemplary aspects may be combined. It is to be understood that the operational steps illustrated in the flowchart diagrams may be subject to numerous different modifications as will be readily apparent to one of skill in the art. Those of skill in the art will also understand that information and signals may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

(61) The previous description of the disclosure is provided to enable any person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other variations. Thus, the disclosure is not intended to be limited to the examples and designs described herein, but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

(62) Implementation examples are also described in the following numbered clauses:

(63) 1. An integrated circuit (IC) package, comprising:

(64) an impedance tuner circuit, comprising: a first switch; and an inductor comprising a first terminal and a second terminal; and a package substrate comprising a first redistribution layer

(RDL), the first RDL comprising the inductor; and a die coupled to the package substrate, the die comprising the first switch; the inductor comprising: a first metal line in the first RDL, the first metal line comprising the first terminal and the second terminal; and a first shunt tap in the first RDL, the first shunt tap coupled to the first terminal and the second terminal; and the first switch coupled to the first shunt tap.
2. The IC package of clause 1, wherein the first switch is configured to be closed to shunt the first shunt tap.
3. The IC package of clause 1 or 2, wherein the first switch is configured to be open to provide an open circuit in the first shunt tap.
4. The IC package of any of clauses 1-3, wherein: the first switch comprises a first switch terminal and a second switch terminal; the first shunt tap comprises: a first shunt metal line coupled to the first terminal; and a second shunt metal line coupled to the second terminal; and the first switch terminal is coupled to the first shunt metal line, and the second switch terminal is coupled to the second shunt metal line.
5. The IC package of clause 4, wherein: the first shunt metal line comprises a first end coupled to the first terminal and a second end; and the second shunt metal line comprises a third end coupled to the second terminal and a fourth end adjacent to the second end.
6. The IC package of any of clauses 1-5, wherein: the first metal line extends along a first longitudinal axis; and the first shunt tap extends along a second longitudinal axis parallel to the first longitudinal axis.
7. The IC package of any of clauses 1-5, wherein: the first metal line extends along a first longitudinal axis; and the first shunt tap extends along a second longitudinal axis orthogonal to the first longitudinal axis.
8. The IC package of any of clauses 1-7, wherein: the impedance tuner circuit further comprises a second switch; and the die further comprises the second switch; the inductor further comprising a second shunt tap coupled to the first terminal and the second terminal; and the second switch coupled to the second shunt tap.
9. The IC package of clause 8, wherein: the second switch comprises a third switch terminal and a fourth switch terminal; the second shunt tap comprises: a third shunt metal line coupled to the first terminal; and a fourth shunt metal line coupled to the second terminal; and the third switch terminal is coupled to the third shunt metal line, and the fourth switch terminal is coupled to the fourth shunt metal line.
10. The IC package of any of clauses 1-10, wherein: the impedance tuner circuit further comprises a second switch; the die further comprises the second switch; the package substrate further comprises a second RDL; and the inductor further comprises: a second shunt tap in the second RDL; a first via coupling the second shunt tap to the first terminal; and a second via coupling the second shunt tap to the second terminal.
11. The IC package of clause 10, wherein: the second switch comprises a third switch terminal and a fourth switch terminal; the second shunt tap comprises: a third shunt metal line coupled to the first terminal; and a fourth shunt metal line coupled to the second terminal; and the third switch terminal is coupled to the third shunt metal line, and the fourth switch terminal is coupled to the fourth shunt metal line.
12. The IC package of clause 10 or 11, wherein: the first metal line extends along a first longitudinal axis; the first shunt tap extends along a second longitudinal axis parallel to the first longitudinal axis; and the second shunt tap extends along a third longitudinal axis that intersects the first longitudinal axis.
13. The IC package of any of clauses 1 to 12, wherein: the die comprises an active semiconductor layer; and the first switch is disposed in the active semiconductor layer.
14. The IC package of clause 4, wherein: the first switch terminal is coupled to a first die interconnect of the die; the second switch terminal is coupled to a second die interconnect of the

die; the first die interconnect is coupled to a first metal interconnect of the package substrate coupled to the first shunt metal line; and the second die interconnect is coupled to a second metal interconnect of the package substrate coupled to the second shunt metal line.
15. The IC package of clause 14, wherein: the package substrate comprises a metallization layer adjacent to the first RDL; and the metallization layer comprises the first metal interconnect and the second metal interconnect.
16. The IC package of any of clauses 1-15 integrated into a device selected from the group consisting of: a set top box; an entertainment unit; a navigation device; a communications device; a fixed location data unit; a mobile location data unit; a global positioning system (GPS) device; a mobile phone; a cellular phone; a smart phone; a session initiation protocol (SIP) phone; a tablet; a phablet; a server, a computer, a portable computer, a mobile computing device; a wearable computing device; a desktop computer; a personal digital assistant (PDA); a monitor; a computer monitor, a television; a tuner, a radio; a satellite radio; a music player; a digital music player; a portable music player, a digital video player; a video player; a digital video disc (DVD) player; a portable digital video player, an automobile; a vehicle component; avionics systems; a drone; and a multicopter.
17. A method of fabricating an integrated circuit (IC) package, comprising: forming a first switch in a die; forming an inductor in a package substrate, comprising: forming a first metal line of the inductor in the first redistribution layer (RDL) in the package substrate, the first metal line comprising a first terminal and a second terminal; and forming a first shunt tap of the inductor in the first RDL, the first shunt tap coupled to the first terminal and the second terminal; coupling the die to the package substrate; and coupling the first switch to the first shunt tap.
18. The method of clause 17, wherein forming the die comprises: disposing an active semiconductor layer on a substrate; and forming the first switch in the active semiconductor layer.
19. The method of clause 17 or 18, further comprising forming the package substrate comprising: forming a first metal interconnect coupled to a first shunt metal line of the first shunt tap; and forming a second metal interconnect coupled to a second shunt metal line of the first shunt tap; and further comprising: coupling a first die interconnect to a first switch terminal of the first switch; coupling a second die interconnect to a second switch terminal of the second switch; and wherein coupling the die to the package substrate comprises: coupling the first die interconnect to the first metal interconnect; and coupling the second die interconnect to the second metal interconnect.
20. The method of clause 19, wherein: forming the package substrate further comprises forming a metallization layer adjacent to the first RDL; forming the first metal interconnect comprises forming the first metal interconnect in the metallization layer; and forming the second metal interconnect comprises forming the second metal interconnect in the metallization layer.
21. The method of any of clauses 17-20, wherein: the first switch comprises a first switch terminal and a second switch terminal; forming the first shunt tap comprises: forming a first shunt metal line in the first RDL coupled to the first terminal; and forming a second shunt metal line in the first RDL coupled to the second terminal; and coupling the die to the package substrate comprises: coupling the first switch terminal to the first shunt metal line; and coupling the second switch terminal to the second shunt metal line.
22. The method of any of clauses 17-21, wherein forming the inductor in the first RDL further comprises forming a second shunt tap in the first RDL, the second shunt tap coupled to the first terminal and second terminal; and further comprising: forming a second switch in the die; and coupling the second switch to the second shunt tap.
23. The method of any of clauses 17-22, further comprising: forming a second switch in the die; and forming the inductor further comprises: forming a second shunt tap in a second RDL in the package substrate, the first shunt tap coupled to the first terminal and the second terminal; forming a first via coupling the second shunt tap to the first terminal; and forming a second via coupling the second shunt tap to the second terminal; and coupling the second switch to the second shunt tap.

Claims

1. An integrated circuit (IC) package, comprising: an impedance tuner circuit, comprising: a first switch; a second switch; and an inductor comprising a first terminal and a second terminal; a package substrate comprising a first redistribution layer (RDL) and a second RDL, the first RDL comprising the inductor; and a die coupled to the package substrate, the die comprising: the first switch; and the second switch; the inductor comprising: a first metal line in the first RDL, the first metal line comprising the first terminal and the second terminal; a first shunt tap in the first RDL, the first shunt tap coupled to the first terminal and the second terminal; a second shunt tap in the second RDL; a first via coupling the second shunt tap to the first terminal; and a second via coupling the second shunt tap to the second terminal; the first switch coupled to the first shunt tap; and the second switch coupled to the second shunt tap.
2. The IC package of claim 1, wherein the first switch is configured to be closed to shunt the first shunt tap.
3. The IC package of claim 1, wherein the first switch is configured to be open to provide an open circuit in the first shunt tap.
4. The IC package of claim 1, wherein: the first switch comprises a first switch terminal and a second switch terminal; the first shunt tap comprises: a first shunt metal line coupled to the first terminal; and a second shunt metal line coupled to the second terminal; and the first switch terminal is coupled to the first shunt metal line, and the second switch terminal is coupled to the second shunt metal line.
5. The IC package of claim 4, wherein: the first shunt metal line comprises a first end coupled to the first terminal and a second end; and the second shunt metal line comprises a third end coupled to the second terminal and a fourth end adjacent to the second end.
6. The IC package of claim 4, wherein: the first switch terminal is coupled to a first die interconnect of the die; the second switch terminal is coupled to a second die interconnect of the die; the first die interconnect is coupled to a first metal interconnect of the package substrate coupled to the first shunt metal line; and the second die interconnect is coupled to a second metal interconnect of the package substrate coupled to the second shunt metal line.
7. The IC package of claim 6, wherein: the package substrate comprises a metallization layer adjacent to the first RDL; and the metallization layer comprises the first metal interconnect and the second metal interconnect.
8. The IC package of claim 1, wherein: the first metal line extends along a first longitudinal axis; and the first shunt tap extends along a second longitudinal axis parallel to the first longitudinal axis.
9. The IC package of claim 1, wherein: the first metal line extends along a first longitudinal axis; and the first shunt tap extends along a second longitudinal axis orthogonal to the first longitudinal axis.
10. The IC package of claim 1, wherein: the second switch comprises a third switch terminal and a fourth switch terminal; the second shunt tap comprises: a third shunt metal line coupled to the first terminal; and a fourth shunt metal line coupled to the second terminal; and the third switch terminal is coupled to the third shunt metal line, and the fourth switch terminal is coupled to the fourth shunt metal line.
11. The IC package of claim 1, wherein: the second switch comprises a third switch terminal and a fourth switch terminal; the second shunt tap comprises: a third shunt metal line coupled to the first terminal; and a fourth shunt metal line coupled to the second terminal; and the third switch terminal is coupled to the third shunt metal line, and the fourth switch terminal is coupled to the fourth shunt metal line.
12. The IC package of claim 1, wherein: the first metal line extends along a first longitudinal axis; the first shunt tap extends along a second longitudinal axis parallel to the first longitudinal axis;

and the second shunt tap extends along a third longitudinal axis that intersects the first longitudinal axis.

13. The IC package of claim 1, wherein: the die comprises an active semiconductor layer; and the first switch is disposed in the active semiconductor layer.

14. The IC package of claim 1 integrated into a device selected from the group consisting of: a set top box; an entertainment unit; a navigation device; a communications device; a fixed location data unit; a mobile location data unit; a global positioning system (GPS) device; a mobile phone; a cellular phone; a smart phone; a session initiation protocol (SIP) phone; a tablet; a phablet; a server, a computer; a portable computer; a mobile computing device; a wearable computing device; a desktop computer; a personal digital assistant (PDA); a monitor; a computer monitor; a television; a tuner; a radio; a satellite radio; a music player; a digital music player; a portable music player; a digital video player; a video player; a digital video disc (DVD) player; a portable digital video player; an automobile; a vehicle component; avionics systems; a drone; and a multicopter.

15. A method of fabricating an integrated circuit (IC) package, comprising: forming a first switch in a die; forming a second switch in the die; forming an inductor in a package substrate, comprising: forming a first metal line of the inductor in a first redistribution layer (RDL) in the package substrate, the first metal line comprising a first terminal and a second terminal; forming a first shunt tap of the inductor in the first RDL, the first shunt tap coupled to the first terminal and the second terminal; forming a second shunt tap in a second RDL in the package substrate, the first shunt tap coupled to the first terminal and the second terminal; forming a first via coupling the second shunt tap to the first terminal; and forming a second via coupling the second shunt tap to the second terminal; coupling the die to the package substrate; coupling the first switch to the first shunt tap; and coupling the second switch to the second shunt tap.

16. The method of claim 15, wherein forming the die comprises: disposing an active semiconductor layer on a substrate; and forming the first switch in the active semiconductor layer.

17. The method of claim 16, further comprising forming the package substrate comprising: forming a first metal interconnect coupled to a first shunt metal line of the first shunt tap; and forming a second metal interconnect coupled to a second shunt metal line of the first shunt tap; and further comprising: coupling a first die interconnect to a first switch terminal of the first switch; coupling a second die interconnect to a second switch terminal of the second switch; and wherein coupling the die to the package substrate comprises: coupling the first die interconnect to the first metal interconnect; and coupling the second die interconnect to the second metal interconnect.

18. The method of claim 17, wherein: forming the package substrate further comprises forming a metallization layer adjacent to the first RDL; forming the first metal interconnect comprises forming the first metal interconnect in the metallization layer; and forming the second metal interconnect comprises forming the second metal interconnect in the metallization layer.

19. The method of claim 15, wherein: the first switch comprises a first switch terminal and a second switch terminal; forming the first shunt tap comprises: forming a first shunt metal line in the first RDL coupled to the first terminal; and forming a second shunt metal line in the first RDL coupled to the second terminal; and coupling the die to the package substrate comprises: coupling the first switch terminal to the first shunt metal line; and coupling the second switch terminal to the second shunt metal line.

20. The method of claim 15, wherein: forming the second switch comprises forming a third switch terminal and a fourth switch terminal in the die; forming the second shunt tap comprises: forming a third shunt metal line in the second RDL coupled to the first terminal; and forming a fourth shunt metal line in the second RDL coupled to the second terminal; and coupling the third switch terminal to the third shunt metal line; and coupling the fourth switch terminal to the fourth shunt metal line.

21. The method of claim 17, wherein: forming the first metal line further comprises forming the first metal line extending along a first longitudinal axis; forming the first shunt tap further

comprises forming the first shunt tap extending along a second longitudinal axis parallel to the first longitudinal axis; and forming the second shunt tap further comprises forming the second shunt tap extending along a third longitudinal axis that intersects the first longitudinal axis.